

Research paper

Optimization techniques for the design of crescent-shaped hard sails for wind-assisted ship propulsion

Cem Guzelbulut^{a,*}, Timoteo Badalotti^b, Katsuyuki Suzuki^a

^a. Department of Systems Innovation, School of Engineering, The University of Tokyo, Japan

^b. Monohakobi Technology Institute, Tokyo, Japan

ARTICLE INFO

Keywords:

Wind-assisted ship propulsion
Rigid windsail
Crescent-shaped profile
Optimization

ABSTRACT

Wind-assisted ship propulsion is one of the most promising technologies to achieve net zero emission in ship operations by 2050 due to its reliance on a fully renewable energy source. Hard sails with crescent profiles have been shown to potentially provide a greater benefit than symmetric sails in recent studies. In order to justify their application, it is necessary to create and adopt a specific design that guarantees a reduction in the engine load over the whole operational range of encounter angles. The current study provides a systematic approach to obtain such a design based on two techniques, one aiming to maximum sail thrust, the other to minimize propeller power. Surrogate models based on validated CFD simulations are adopted for optimization, and the most efficient design is found as an average over all the possible encounter angles. It was shown that the thrust maximized approach increased the average thrust generated by sail up to 12.3%, and the propeller power minimized approach improved the effectiveness of the sails up to 22% in terms of percent power reduction. It was also confirmed that crescent-shaped airfoils generally outperform symmetric airfoils in the considered conditions.

1. Introduction

The shipping industry, which is fundamental for today's trade network and material supply chain, has undergone significant changes in terms of the design, operation and management of ships due to the high portion of the global CO₂ emission. In particular, the regulations proposed by the International Maritime Organization aim to achieve net-zero CO₂ emissions from the maritime industry by 2050 (IMO, 2023). Therefore, new tools to reduce fuel consumption of ships to be employed at the design and operational stage are needed, such as route-planning tools, carbon capture and storage technologies, wind propulsion.

The most straightforward solution to decarbonize the sector consists of switching from heavy marine oil to alternative clean fuels. Within this framework, Law et al. (2021) compared different pathways considering different fuel types in the shipping industry. It was found that carbon capture and storage systems are promising technology, but it is still unclear how to deal with accumulated CO₂. Alternative fuels have been found to present issues related to technological readiness level, fuel cost, and NOx emissions. A study conducted by Tomos et al. (2024) investigated the lifecycle of alternative fuels applicable to the shipping

industry, and the key result emphasized that none of the alternative fuel alone is adequate to achieve the zero-emission goal and multiple solutions should be utilized simultaneously. Based on such earlier studies, it can be concluded that new fuels are not enough, but additional reduction of ship resistance, improvement of the propulsion characteristics, and auxiliary propulsion systems should be incorporated to achieve net-zero GHG-emissions.

As regards the resistance of ships, reduction is possible by optimizing the hull shape, installing air lubrication systems, reducing the ship speed, etc. Park et al. (2015) optimized the shape of bow and stern separately to minimize the viscous and wave making resistance of the hull form and an optimal hull form was obtained reducing the required power by 3.7%. In addition to reducing the ship resistance, the propulsion characteristics of the propeller can be improved by Tadros et al. (2021, 2022).

Interest in wind sails, which had given way to steam engines first and later internal combustion engines several decades ago, was recently rekindled due to their use of wind as a fully renewable energy source. Many different systems have been proposed and have been developed to harness the wind energy available during ocean transit. In particular, rigid wind sails generate lift and drag forces of varying magnitudes depending on the airfoil geometry, and their angle of attack can be

* Corresponding author. The University of Tokyo, 7-3-1 Hongo, Bunkyo-ku, Tokyo, 113- 8656, Japan.

E-mail address: cem@struct.t.u-tokyo.ac.jp (C. Guzelbulut).

<https://doi.org/10.1016/j.oceaneng.2024.119142>

Received 12 June 2024; Received in revised form 22 August 2024; Accepted 30 August 2024

Available online 3 September 2024

0029-8018/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

Nomenclature			
A :	Projected sail area [m ²]	N_R :	Moment generated by rudder in yaw direction [N·m]
B :	Ship breadth [m]	N_{WAD} :	Moment generated by sail in yaw direction [N·m]
C_B :	Block coefficient	P :	Pressure [Pa]
C_D :	Drag coefficient	P_{prop} :	Change in average propeller power compared to no-sail case [%]
$C_{D,max}$:	The maximum drag coefficient	P_∞ :	Ambient pressure [Pa]
$C_{D,min}$:	The minimum drag coefficient	r :	Ship speed in yaw direction [rad/s]
c_i :	The coefficient vector used in the lift/drag model	R_1 :	The radius of inscribed circle at mid-chord [m]
C_L :	Lift coefficient	R_2 :	The edge radius on both edges of crescent airfoil [m]
$c_{L,D}$:	The array of lift and drag coefficients at different attack angles	R_3 :	The radius of curvature of the mean airfoil camber of crescent airfoil [m]
$C_{L,max}$:	The maximum lift coefficient	T :	Sail thrust [N]
$C_{L,0}$:	Lift coefficient for zero angle of attack	TWD :	True wind direction [°]
C_P :	Pressure coefficient	TWS :	True wind speed [m/s]
D :	Drag force [N]	u :	Ship speed in surge direction [m/s]
D_P :	The diameter of propeller [m]	V_a :	Apparent wind speed [m/s]
d :	Ship draft [m]	v_m :	Ship speed in sway direction at mid-ship [m/s]
d_i :	Propeller power percentage coefficient	X :	Total force in surge direction [N]
I_{zG} :	Inertia of ship in yaw direction [kg·m ²]	x_G :	The position of center of mass with respect to mid-ship [m]
J_z :	Added inertia of ship in yaw direction [kg·m ²]	X_H :	Hull hydrodynamic force in surge direction [N]
J'_z :	Normalized added inertia of ship in yaw direction [kg·m ²]	$X_{Hull-Wave}$:	Hull-wave interaction force in surge direction [N]
L :	Lift force [N]	$X_{Hull-Wind}$:	Hull-wind interaction force in surge direction [N]
L_{pp} :	The length of the ship between perpendiculars [m]	X_P :	Propeller force in surge direction [N]
m :	Mass of the ship [kg]	X_R :	Rudder force in surge direction [N]
m_x :	Added mass of the ship in surge direction [kg]	X_{WAD} :	Sail force in surge direction [N]
m'_x :	Normalized added mass of the ship in surge direction	Y :	Total force in sway direction [N]
m_y :	Added mass of the ship in sway direction [kg]	Y_H :	Hull hydrodynamic force in sway direction [N]
m'_y :	Normalized added mass of the ship in sway direction	$Y_{Hull-Wave}$:	Hull-wave interaction force in sway direction [N]
N :	Total moment in yaw direction [N·m]	$Y_{Hull-Wind}$:	Hull-wind interaction force in sway direction [N]
n_{cases} :	The number of environmental cases	Y_P :	Propeller force in sway direction [N]
N_H :	Moment generated by hull hydrodynamic in yaw direction [N·m]	Y_R :	Rudder force in sway direction [N]
$N_{Hull-Wave}$:	Moment generated by hull-wave interaction in yaw direction [N·m]	Y_{WAD} :	Sail force in sway direction [N]
$N_{Hull-Wind}$:	Moment generated by hull-wind interaction in yaw direction [N·m]	α :	Angle of attack [°]
N_P :	Moment generated by propeller in yaw direction [N·m]	$\alpha_{L,max}$:	Angle of attack for the maximum lift [°]
		α_{max} :	Angle of attack for the maximum thrust [°]
		∇ :	Displacement volume [m ³]
		ρ :	The density of air [kg/m ³]

adjusted depending on the wind direction to provide maximum thrust to the ship. Alternatively, rotor sails, in their simplest form, are rotating cylinders fitted on top of the ship and exploiting the Magnus effect, where their rotational speed can be set to maximize the thrust, they generate depending on the direction and speed of the incoming wind. Tillig and Ringsberg (2019) studied the effects of different numbers and arrangements of rotor sails on different ships and showed that reductions in fuel consumption of up to 30% could be achieved. Guzelbulut and Suzuki (2023) optimized the design of single rotor sail taking into account the required capital and fuel consumption and showed that the optimal rotor sail for the KVLCC2 ship can reduce energy consumption by about 4%. Kites represent another alternative solution. Due to stronger wind conditions at higher altitudes, kites can generate significant thrust forces for backwind conditions. Formosa et al. (2023) investigated static kite sails and found that static kites with a large surface area can provide the total required thrust at high wind speeds in slow-steaming conditions.

Among the different types of wind sails, rigid wind sails are the focus of the present study. Previous studies on rigid wind sails have focused on symmetric NACA airfoils, multi-element wind sails, blowing-suction type airfoils and crescent shaped airfoils. Reche-Vilanova et al. (2021) compared wind-assistance systems, including rigid wind sails made of symmetric solid airfoils and airfoils with flaps, and showed that airfoils

with flaps have better performance in terms of power savings because of their additional flexibility afforded by the angle of the flap. Viola et al. (2015) proposed a two-element wing combining the NACA0012 and NACA0010 airfoils. The effect of the proposed two-element wing with different area and aspect ratios of sails on power savings was investigated in different wind conditions and at different cruising speeds, and it was estimated that power savings of up to 10% could be achieved. In another study, Lee et al. (2016) combined three identical NACA0012 airfoils and analyzed the effects of flap and interaction between sails. They found that the power savings per single sail decreased when multiple sails were used and that the use of flaps increased the power savings. In another study conducted by Cairns et al. (2021) to improve the performance of symmetric NACA airfoils, a blowing-suction type airfoil with a circulation control system was proposed, and it was found that the maximum lift coefficient increases when such a system is employed. Another type of airfoil used in rigid wind sails is the crescent shaped airfoil. Zhu et al. (2023) investigated the lift and drag characteristics of a crescent shaped airfoil and observed that the crescent shape generates significantly larger thrust force compared to a NACA0015 airfoil. Nikmanesh (2021) compared four different crescent-shaped airfoils through CFD analysis and concluded that increasing curvature and decreasing thickness have a positive effect on the thrust. Because crescent-shaped airfoils feature better thrust generation characteristics

than symmetric NACA airfoils while being less mechanically complex than airfoils fitted with flaps or with suction mechanisms, they were chosen as the subject of further investigation in the present study. Many studies were conducted to understand the effect of different wind-assisted propulsion systems on energy consumption and the interactions between sails, however, the aim of the present study is to develop an optimization-based design methodology to determine the best airfoil shape considering the target ship and route.

In the current investigation, it was aimed to obtain the optimal crescent shape that maximizes thrust and minimizes propeller power. In the ‘Methods’ section, the parameters used to define the crescent airfoil were introduced. Then, a two-dimensional CFD model of a crescent-shaped airfoil was generated, and the results were compared to previous studies for validation of the simulation settings. After ensuring the accuracy of CFD analysis, a set of example airfoils were created based on the Taguchi orthogonal-array method. CFD analysis of each airfoil was conducted for angles of attack within the range of 0–90°. Subsequently, two different optimization schemes were considered: to maximize the average thrust generated over the whole range of true-wind angles with respect to the ship, where the angle of attack is chosen to maximize thrust for each true-wind angle; to minimize the average propeller power for true wind angles between 0 and 180°, where the angle of attack is chosen to minimize power for each true-wind angle. In order to implement the former approach, a surrogate model for lift and drag coefficients depending on the design parameters of the airfoil was created. For the latter approach, a ship dynamics model, namely the MMG model (Ogawa et al., 1977; Yasukawa and Yoshimura, 2015), was used to find the propeller power by maintaining the ship speed on a given course, and a surrogate model for the average propeller power was generated depending on the design parameters of the airfoil. In the sections of ‘Results’ and ‘Discussion’, the performances of both the optimal results are compared to understand how much weight the ship dynamics has on the optimization and how each design parameter affects the performance. Finally, a comparison between optimal crescent shaped airfoils and symmetric NACA airfoils to extend our understanding of both types of airfoils is presented. The findings of the study, limitations and future studies were given in the ‘Conclusion’ section.

2. Methods

2.1. Description of wing-sail shape and aerodynamic parameters

In the present study, a telescopic rigid wind sail featuring a crescent-shaped profile was considered. The airfoil profile is defined by three parameters: airfoil thickness, which is defined in terms of the radius of an inscribed circle at mid-chord, R_1 , edge radius of curvature, which is equal for both edges, R_2 , and radius of curvature of the airfoil mean camber, R_3 , as shown in Fig. 1. Additionally, the upper and lower circular arcs of the airfoil are tangent to both circles on edges of radius R_2

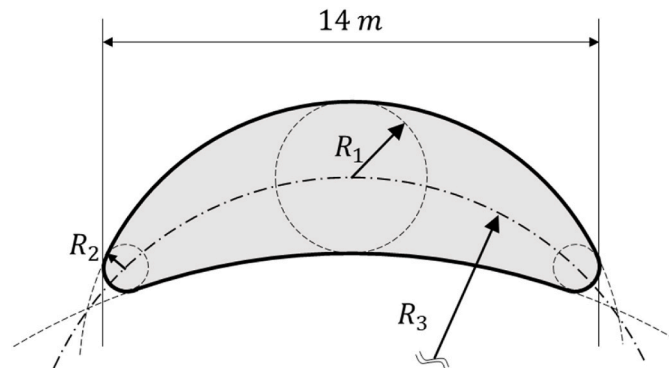


Fig. 1. The definition of design variables of a crescent shaped airfoil having a width of 14 m.

and the central circle of radius R_1 .

The height of the sail and the chord of the airfoil were fixed at 50 m and 14 m, respectively, in order to compare the numerical results herein obtained with relevant studies available in the literature (Zhu et al., 2023) (Nikmanesh, 2021), where the wing sail parameters will be provided in non-dimensional format, as in Eqs. (1)–(3), for airfoil shape comparison as follows:

$$C_p = \frac{P - P_\infty}{\frac{1}{2} \rho V_a^2} \quad (1)$$

$$C_L = \frac{L}{\frac{1}{2} \rho V_a^2 A} \quad (2)$$

$$C_D = \frac{D}{\frac{1}{2} \rho V_a^2 A} \quad (3)$$

2.2. Generation a series of airfoils based on Taguchi orthogonal arrays

To understand how each design variable affects the aerodynamics of the airfoils and to optimize the performance of sails, a design of experiments study was conducted. Taguchi orthogonal arrays were used to determine a set of designs of crescent airfoils. (Roy RK, 1990).

Taguchi's orthogonal arrays are a methodology for standardizing Design of Experiments aimed at minimizing a loss function. A number of levels is specified, which represents the cardinality of the set of values each parameter may assume, that is, each parameter is assigned a given value for each level, where the number of levels is the same for all parameters. For the analysis in question, three levels were adopted, leading to three different values that R_1 , R_2 , and R_3 may take on as given in Table 1. The upper and lower bounds of design variables were determined to avoid largely cambered airfoils and keep crescent-like shape. An orthogonal array is then defined, where each row represents a trial, i. e. an experiment, with the level of each parameter in the trial assigned in the corresponding column as shown in Table 2. Fig. 2 shows the resulting airfoil profile for each of the nine combinations of parameters.

The array is said to be orthogonal in that: all the columns have an equal number of levels; taken any two columns, they provide the same number of combinations of parameter levels. To exemplify the latter, in Table 2, (1,1), (1,2), (1,3), (2,1), and so on occur each once in columns one and two (R_1 Level, R_2 Level) as they do in columns two and three (R_2 Level, R_3 Level). For a three-parameter three-level study, nine is the minimum number of trials leading to an orthogonal array.

The advantage of the Taguchi method is that it provides a comprehensive methodology to study the effect of the variation of each parameter on a loss function with a number of trials that is less than that required by a full factorial. For the case at hand, a full factorial would require $3^3 = 27$ trials as opposed to nine. This facilitates sensitivity analysis and optimization.

2.3. Aerodynamic characteristics of airfoils determined by Taguchi method

In order to investigate the performance of airfoils, computational fluid dynamics (CFD) simulations were conducted using Altair Hypermesh CFD 2023 (Altair Hypermesh CFD Software, 2023) (mesher) and AcuSolve 2023 (Altair AcuSolve Software, 2023) (solver). The solver is based on a Finite Volume Method, and a steady implementation of the

Table 1
The value of each level for design variable.

Parameter	Level 1	Level 2	Level 3
R_1 (m)	0.5	1	1.5
R_2 (m)	0.2	0.4	0.6
R_3 (m)	8	13	18

Table 2

The level of each design variable for a set of airfoil designs determined by the Taguchi method.

Airfoil Design	R ₁ Level	R ₂ Level	R ₃ Level
#1	1	1	1
#2	1	2	2
#3	1	3	3
#4	2	1	2
#5	2	2	3
#6	2	3	1
#7	3	1	3
#8	3	2	1
#9	3	3	2

solver was adopted to limit computational time. Acting forces on the airfoil were predicted considering different angles of attacks: 0, 10, 15, 20, 25, 30, 40, 50, 60, 70, 80, 90°. As reported in a previous study (Zhu et al., 2023), the critical angle of attack is expected to be around 20° and the number of datapoints were increased in its vicinity.

The flow around the airfoil was assumed to be two-dimensional and steady, and the fluid to be incompressible air with the following prop-

erties: density of 1.225 kg/m³ and dynamic viscosity of 1.781×10^{-5} kg/m.s. The inflow velocity is 25 m/s, which results in a Reynolds number of approximately 2.4×10^7 .

The airfoil was placed approximately 10 times the chord length downstream of the inlet and away from the side boundaries of the fluid domain as shown in Fig. 3. No-slip boundary conditions were assigned to the airfoil profile. Symmetric boundary conditions were imposed on the opposite sides of extruded mesh. The sides and downstream boundaries were assigned pressure-outlet boundary conditions.

The domain of the problem was discretized at first through a triangular surface mesh with the following feature as shown in Fig. 4.

- Maximum mesh size of 10 m
- Near-field refinement with cell size of 1.12 m in a bounding box of length of 150 m and height of 80 m starting 40 m upstream of the profile.
- Near-field refinement with cell size of 0.28 m in a bounding box of length of 100 m and height of 40 m starting 20 m upstream of the profile.
- A prism layer with a thickness of 0.21 m.

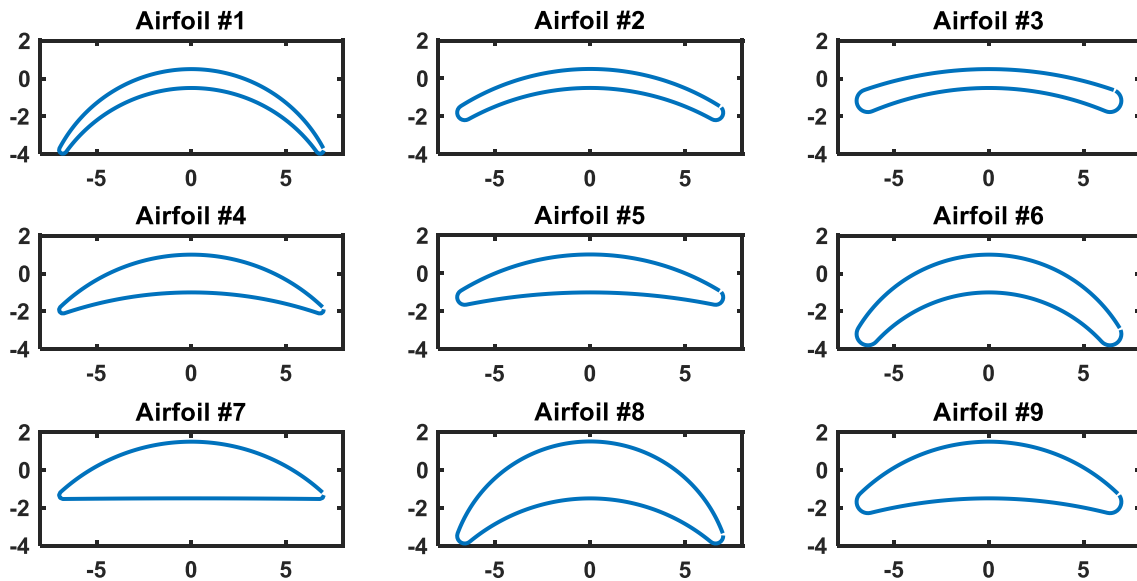


Fig. 2. Design of airfoil profiles generated by Taguchi method.

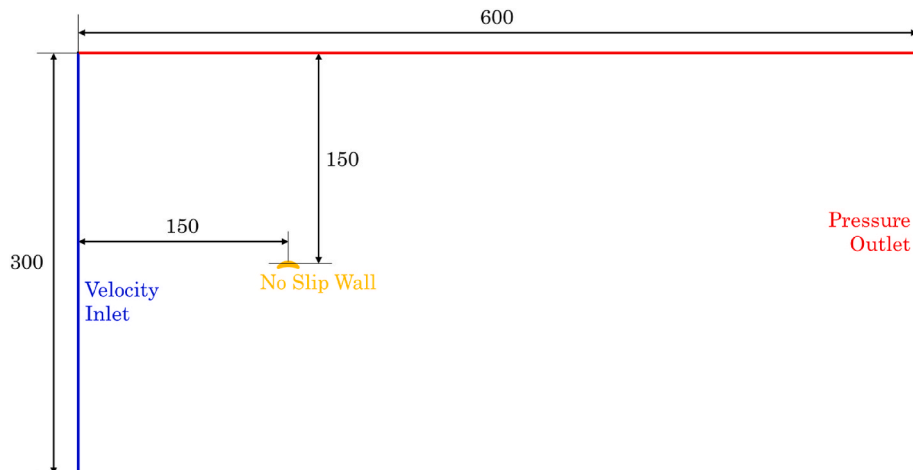


Fig. 3. Boundary conditions assigned to model.

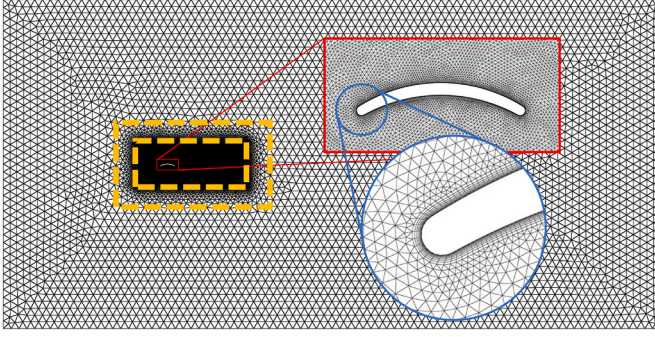


Fig. 4. Mesh refinements used in CFD model.

- The average number of nodes, volume meshes, hexahedral elements and prism elements are 138600, 135000, 3150, and 13850, respectively.

Suitability of the settings above was checked by confirming that the settings provide solutions consistent with the literature.

Then, triangular elements were extruded, and one element thick prism elements were created. The flow field was modeled by steady incompressible RANS (Reynolds-Average Navier Stokes equations), and an SST $k-\omega$ model was adopted for turbulence.

2.4. Optimization of the shape of crescent airfoil

Two different approaches were considered to find out the optimal shape of crescent airfoils. In the first approach, only the wind sail system was considered. Lift and drag coefficients were calculated for the 0–90-degree range of angle of attack and a column vector, $c_{L,D}$, which includes lift coefficients in the upper half and drag coefficients in the lower half in ascending order of angle of attack, was assembled. Then, second order polynomials were used to predict the effect of airfoil design on lift and drag coefficients between level values of the parameters as previously detailed, as in Eq. (4). The coefficients are listed in Appendix A.

$$c_{L,D} = c_1 + c_2 \times R_1 + c_3 \times R_2 + c_4 \times R_3 + c_5 \times R_1^2 + c_6 \times R_2^2 + c_7 \times R_3^2 \quad (4)$$

When wind sails are fitted on a ship and the force generated by sails are calculated based on the apparent wind speed, apparent wind direction and angle of attack. The apparent wind speed and angle are determined by the true wind speed, true wind direction, and ship speed. The angle of attack of the airfoil is the angle between chord and flow direction. Ship speed and true wind speed were assumed to be 8 and 15 m/s, respectively. The true wind direction was set from 0° to 180° with 30-degree steps. Then, the angle of attack, α , was optimized to maximize the thrust force during the simulation of each case as shown in Eq. (5), where the thrust is denoted by T which depends on true wind speed, TWS_i , true wind direction, TWD_i , and lift and drag coefficients, C_L and C_D , which are determined by the design parameters, R_1 , R_2 and R_3 , and the angle of attack, α . The angle of attack which gives the maximum thrust for given shape and environmental condition is defined as α_{max} . The optimal shape of the airfoil was found by maximizing the average thrust generated by the sail over environmental cases, n_{cases} , as in Eq. (6). It should be pointed out that the airfoil lift and drag coefficients were obtained for a flow speed of 25 m/s, but no significant change in behavior is expected in the range of application.

$$\max_{\alpha} T \left(TWS_i, TWD_i, C_{L(R_1, R_2, R_3, \alpha)}, C_{D(R_1, R_2, R_3, \alpha)} \right) \quad (5)$$

subject to $0 \leq \alpha \leq 90$

$$\max_{R_1, R_2, R_3} \frac{\sum_{i=1}^{n_{cases}} T \left(TWS_i, TWD_i, C_{L(R_1, R_2, R_3, \alpha_{max})}, C_{D(R_1, R_2, R_3, \alpha_{max})} \right)}{n_{cases}} \quad (6)$$

In the second approach, a ship dynamics model, known as MMG model, was used to predict how the crescent shape airfoil affects the power consumption of the ship. The equation of motion of the ship is given based on the MMG model approach, see Eqs. (7) and (8). In the MMG model, each force component is defined independently. A detailed investigation of the MMG model for wind-assisted ships can be found in previous work (Guzelbulut et al., 2024). Lift and drag coefficients of the base airfoils determined by the Taguchi array were implemented on the MMG model of the target ship KVLCC2 (MOERI, 2023). The main particulars of the target ship, KVLCC2, are given in Table 3. Further details and parameters related to each force component can be found in (Yasukawa and Yoshimura, 2015) and (Guzelbulut et al., 2024).

$$\begin{aligned} (m + m_x)\ddot{u} - (m + m_y)v_m r - x_G m \dot{r}^2 &= X \\ (m + m_y)\ddot{v}_m + (m + m_x)u r + x_G m \dot{r} &= Y \\ (I_{zG} + x_G^2 m + J_{zz})\ddot{r} + x_G m(\dot{v}_m + u r) &= N_m \end{aligned} \quad (7)$$

$$\begin{aligned} X &= X_H + X_P + X_R + X_{Hull-Wind} + X_{Hull-Wave} + X_{WAD} \\ Y &= Y_H + Y_P + Y_R + Y_{Hull-Wind} + Y_{Hull-Wave} + Y_{WAD} \\ N_m &= N_H + N_P + N_R + N_{Hull-Wind} + N_{Hull-Wave} + N_{WAD} \end{aligned} \quad (8)$$

The propeller power was calculated considering a true wind speed of 15 m/s at different true wind directions for a ship travelling at 8 m/s. Then, the average propeller power was calculated for each airfoil oriented at the angle leading to the minimum propeller power, and a second order polynomial was used to express the relation between the design parameters of the airfoil and change in average propeller power in percentage considering different environmental cases in Eq. (9). Using this relation between airfoil design and propeller power, the airfoil design is optimized to minimize the average propeller power, as given in Eq. (10). The coefficients were listed in Appendix B.

$$P_{prop} = d_1 + d_2 \times R_1 + d_3 \times R_2 + d_4 \times R_3 + d_5 \times R_1^2 + d_6 \times R_2^2 + d_7 \times R_3^2 \quad (9)$$

$$\min_{R_1, R_2, R_3} P_{prop}(R_1, R_2, R_3) \quad (10)$$

The optimization problems were solved using MATLAB R2022b, Global Optimization Toolbox, “surrogateopt” function.

3. Results

In the present study, two-dimensional steady flow assumptions were adopted to predict the aerodynamics of crescent shaped airfoils. For this purpose, CFD analyses were conducted for the same crescent airfoil used in (Zhu et al., 2023), and the results obtained were compared with the reference study, which employed both two-dimensional and three-dimensional unsteady RANS, with the same turbulence model as

Table 3

The main particulars of the target ship, KVLCC2.

Parameter	Value
L_{pp} (m)	320
B (m)	58
d (m)	20.8
D_p (m)	9.86
∇ (m ³)	312,600
x_G (m)	11.2
C_B	0.81
m'_x	0.022
m'_y	0.223
J_z	0.011

ours. The design parameters, R_1 , R_2 , and R_3 , of the reference airfoil used in (Zhu et al., 2023) are 1 m, 0.2 m, and 12.812 m, respectively.

The comparison was carried out in terms of pressure coefficient, c_p , lift coefficient, c_L , and drag coefficient, c_D for validation purposes. The pressure coefficient, c_p , along the chord from the present and the reference study are given for angles of attack of 10, 20, 60 and 90° in Fig. 5. The pressure distribution on the pressure side of the airfoil shows excellent agreement between present and reference predictions both in two-dimensional and three-dimensional, where the only discrepancy can be seen for a very high angle of attack, 90°, that is airfoil transverse to the flow, at the edges, where the present method gives consistent results with the reference three-dimensional method but not the reference two-dimensional method. As to the suction side, the present method generally agrees with the reference values for two-dimensional and three-dimensional results for low angles of attack, while, for large angles of attack, large variations are observed in the reference between two-dimensional and three-dimensional results, and the present method agrees mostly with the reference three-dimensional results. Therefore, where separation and ensuing unsteadiness is not dominant (pressure side for low angles of attack or high angles of attack far from the edges, suction side at low angles of attack), all the methods give similar results, while, where separation is pronounced, the different methods give different predictions, and the present method agrees better with reference three-dimensional predictions than reference two-dimensional predictions. The lift and drag coefficients of the reference airfoil were also compared to those from the previous study (Zhu et al., 2023) to further ensure simulation settings were adequate. It was found that two-dimensional CFD simulations in the present study resulted in very similar results to three-dimensional CFD simulations, as shown in Fig. 6. Provided that the reference study eventually adopts two-dimensional simulations for the analysis with a scaling factor to account for differences between two-dimensional and three-dimensional simulations in order to reduce the computational cost, as long as the present method

agrees with the three-dimensional reference values, it is expected that agreement is satisfactory for a shape optimization meant to determine the parameters R_1 , R_2 , and R_3 at the design stage. Eventually, at the detailed design stage, this shape will be the base shape for further three-dimensional analysis and refinement of the shape minor details, with possible adjustment of R_1 , R_2 , and R_3 in the vicinity of their current values. This analysis will also consider the shape of the ship superstructure.

Then, lift and drag characteristics of all the Taguchi trials of crescent shaped airfoils were investigated using the same simulation settings. Lift and drag coefficients, c_L and c_D , were obtained for the angles of attack of 0, 10, 15, 20, 25, 30, 40, 50, 60, 70, 80 and 90° and splines were fitted as shown in Fig. 7. The following can be observed:

- The maximum lift and drag coefficients, $c_{L,max}$ and $c_{D,max}$, varied within a range of 1.5–1.85 and 1.25–1.85, respectively.
- The angle of attack giving maximum lift, $\alpha_{L,max}$, was in the range of 15–30°.
- The lift coefficient at zero-degree angle of attack, $c_{L,0}$, is not zero, as would be the case for a symmetric NACA profile.
- Drag coefficient, c_D , was significantly higher than that of symmetric NACA airfoils before stall condition, which has approximately 15–20 degrees of angle of attack.

The first approach to improve the design of crescent airfoils was to focus on maximizing the average thrust power generated by sails by neglecting the ship dynamics. When calculating average thrust power, angle of attack is decided by maximizing the thrust force for each condition of true wind angle. Thus, the optimal design, which operates at optimal angle of attack depending on wind direction, was found as shown in Fig. 8. The design parameters of thrust-maximized airfoil were found as $R_1 = 0.5$ m, $R_2 = 0.2$ m, and $R_3 = 9.2633$ m. The optimal design has more camber but less thickness in the mid-chord. The average thrust

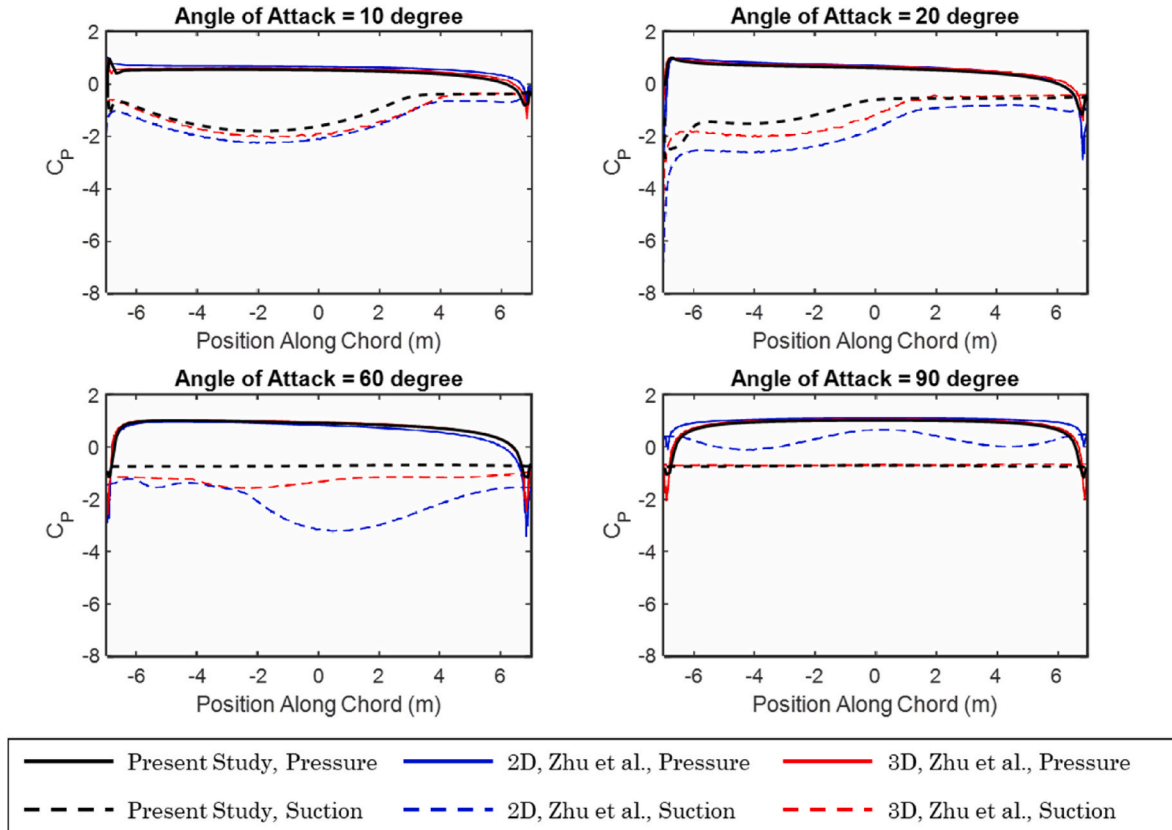


Fig. 5. The comparison of pressure coefficient, c_p , over the chord at different angles of attack.

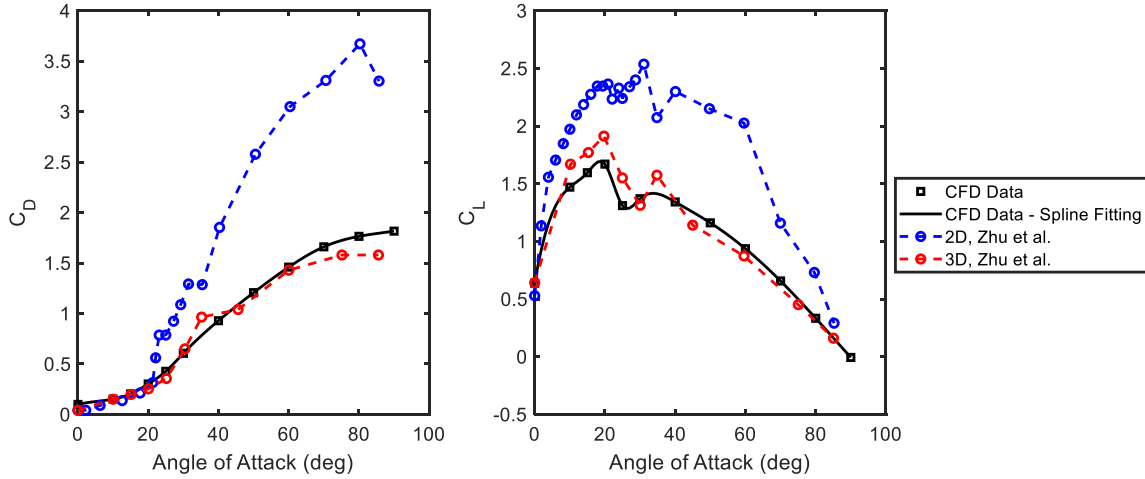


Fig. 6. Comparison of lift and drag coefficients, c_L and c_D , for the angles of attack between 0 and 90°.

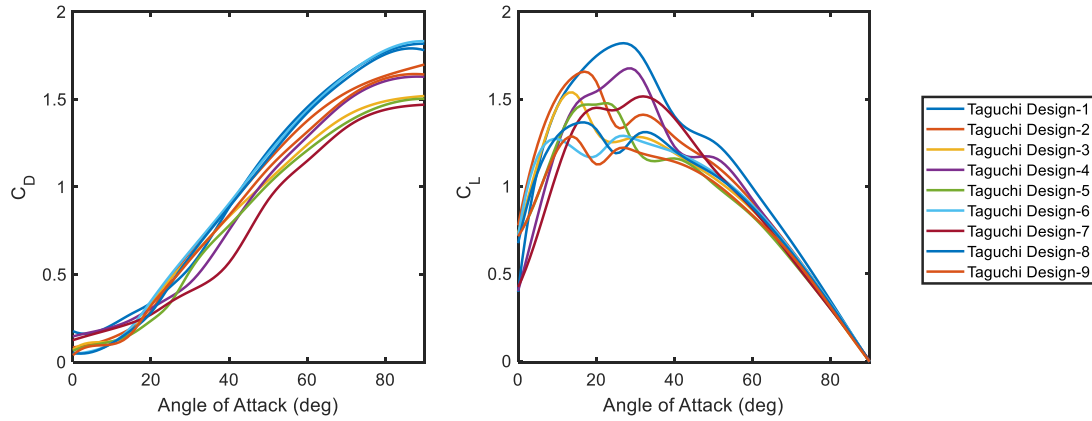


Fig. 7. Variation of lift and drag coefficients, c_L and c_D , depending on the design of airfoil.

power was about 12.3% larger than that of reference airfoil. The maximum lift coefficient of the optimal design, $c_{L,max}$, was found to be 1.97, which is approximately 15% larger than that of the reference airfoil. However, the drag coefficient showed little variation according to reference airfoil. In terms of thrust power, the optimal design can generate up to 23% larger thrust than the reference airfoil. However, for head winds, it generates larger breaking power due to a larger minimum drag coefficient, $c_{D,min}$, caused by larger camber.

In the second approach, the ship dynamics was also taken into account and the design of the crescent shaped airfoil was optimized to minimize propeller power. The design parameters of power-minimized airfoil were found to be $R_1 = 0.5$ m, $R_2 = 0.2$ m, and $R_3 = 8$ m. The optimal design, which minimizes propeller power, further improved the maximum lift coefficient to 2.14, which is approximately 26% larger than the reference airfoil, as shown in Fig. 9. Analogously to thrust-maximized airfoil, the power-minimized airfoil did not show significant differences in terms of drag coefficients either. Then, the power reduction characteristics of reference airfoil, thrust-maximized airfoil and power-minimized airfoil were calculated by implementing the aerodynamics of each airfoil into MMG model after recomputing CFD for the optimal shapes obtained from both methods. To quantify the benefits of this profile, for a sail system having an area of 700 m², the power-minimized design improves the effectiveness of sails up to 5.5% and 22% compared to the thrust-minimized airfoil and the reference airfoil in terms of percent power reduction.

Another important topic to be considered is how different crescent

shaped airfoils and different symmetric NACA airfoils behave and what kind of variations exist in terms of aerodynamic performance. For this reason, the reference crescent-shaped airfoil, thrust-maximized design, power-minimized design, NACA0012 and NACA0025 were compared in terms of lift and drag coefficients for different angles of attack, in terms of power, where the power comparison takes ships dynamics into consideration for all shapes, and in terms of optimal angles of attack under different true wind angles, that is in operative wind conditions for a wind flowing toward the ship. Fig. 10 shows the results.

For small angles of attack, the crescent shaped airfoils have significantly larger lift and drag coefficients. Additionally, the maximum lift coefficient of crescent shaped airfoils was found to be larger than of symmetric NACA airfoils. Consequently, in operative conditions, where the optimal angle of attack is small for head winds, symmetric NACA airfoils, having a lower drag coefficient, performed slightly better. However, for beam and quartering winds, which determine the largest portion of resistance, crescent-shaped airfoils generate a significantly larger maximum lift coefficient, which makes them a better choice altogether, as shown in Fig. 10(c–d). Another important outcome was crescent shaped airfoils use lift-based thrust generation more than symmetric NACA airfoils, as shown in Fig. 10(d).

4. Discussion

The present study aimed to understand the effects of the design parameters of crescent-shaped airfoils on the reduction of required

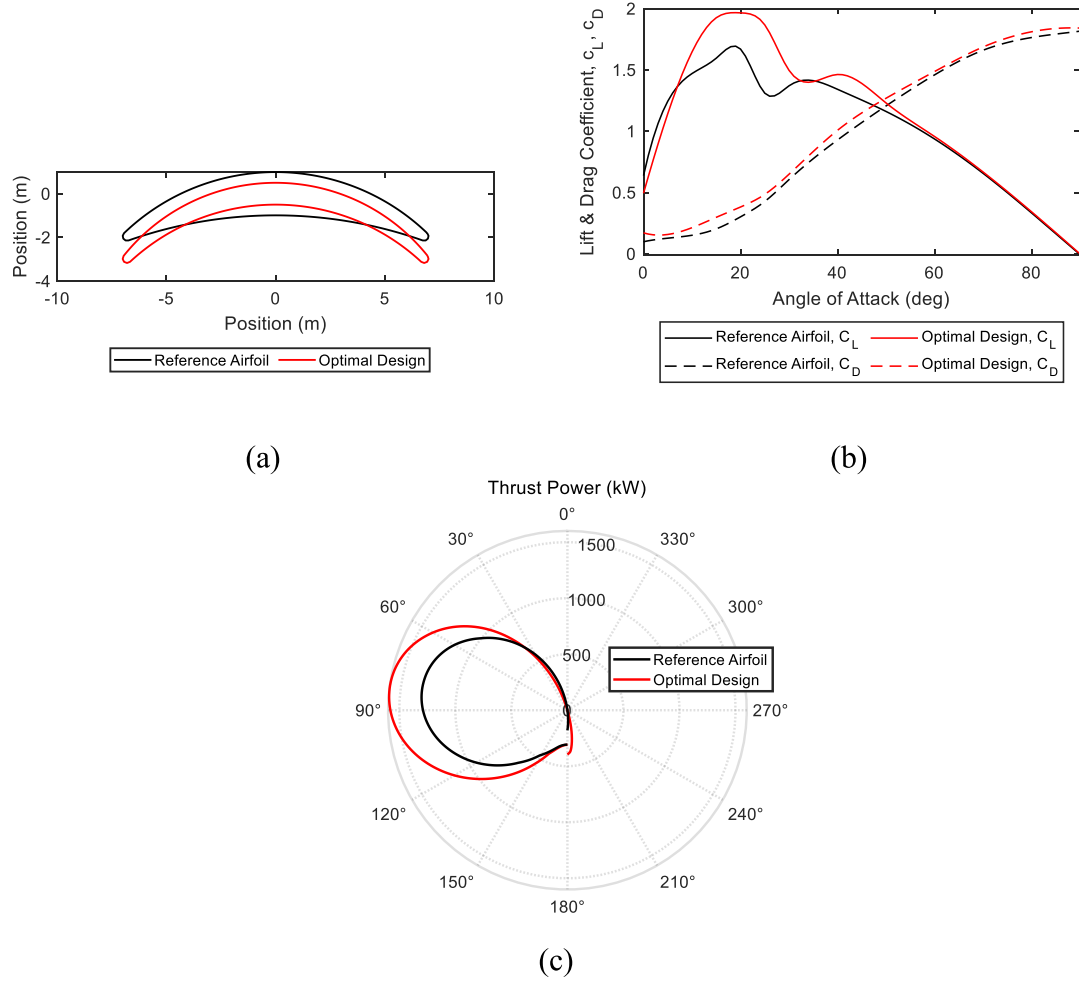


Fig. 8. The comparison between thrust-maximized design and reference airfoil in terms of (a) geometry, (b) lift and drag coefficients, (c) thrust power generated at different true wind angles.

propulsive power and to optimize the airfoil shape to increase thrust and power savings. In order to evaluate the performance of crescent-shaped airfoils, CFD simulations were conducted at different angles of attack. Surrogate models for lift/drag coefficients and propeller power were created. Then, the thrust-maximized and power-minimized designs of the airfoil were found.

Recent CFD studies used both two-dimensional and three-dimensional flow-field models to solve the unsteady RANS set of equations. For the sake of simplicity, two-dimensional and steady RANS simulations were considered in the present study. According to Figs. 5 and 6, the two-dimensional CFD simulations used in the present study showed good agreement with the two-dimensional and three-dimensional averaged unsteady RANS simulations used in the previous studies. Although some differences exist between the two-dimensional steady simulations in the present study and the three-dimensional, time-averaged, unsteady simulations in the previous study (Zhu et al., 2023), the discrepancy was found to be less than that between the two-dimensional and three-dimensional simulations in the previous study. Provided that a scaling factor between the two-dimensional and three-dimensional simulations was employed in the previous study, it can be said that simulations conducted in a two-dimensional domain are accurate enough to predict the relative effect of each parameter on the airfoil performance, which accuracy is high enough for design optimization, while absolute values require a three-dimensional assessment.

It should be noted that airfoils perform best (high lift and low drag) at small angles of attack. When the apparent wind direction is 90 deg,

the airfoil is roughly oriented with its chord perpendicular to the ship longitudinal, where high lift is directed along the longitudinal, thus generating the most thrust while drag is relatively small if compared with other angles of attack and its contribution to sway is minimal.

The design parameters defining the crescent-shaped airfoil were determined as center line radius, thickness at mid-chord, and edge radius. It was found that reducing the center line radius, in other words increasing the camber, and reducing thickness and edge radius improves the performance of sails. Additionally, it was observed that an increment in camber leads to an increase in the maximum drag coefficient, $c_{D,max}$, as shown in Figs. 2 and 7. Moreover, reducing the thickness and edge radius brings forth an increase in the maximum lift coefficient, $c_{L,max}$.

When the thrust-maximized and the power-minimized approaches were compared to find the optimal shape of the airfoil, the latter was found to be superior to the former. The main reason for this is that side forces generated by sails are not included in the former approach. Side forces generated by sails cause rudder angle to change slightly in order to follow the given route and ship moves with drift, which contributes to the ship resistance. The second reason is that the irregularity of the aerodynamic coefficients in the vicinity of the stall region poses challenges to build a surrogate model of lift and drag coefficients in the thrust maximization approach, while the propeller power minimization approach pre-selects the optimal angle of attack in the region with maximum thrust without the need to have a surrogate model over the whole range of angles of attack. Therefore, because of this difference, the algorithm on which the power minimization approach is

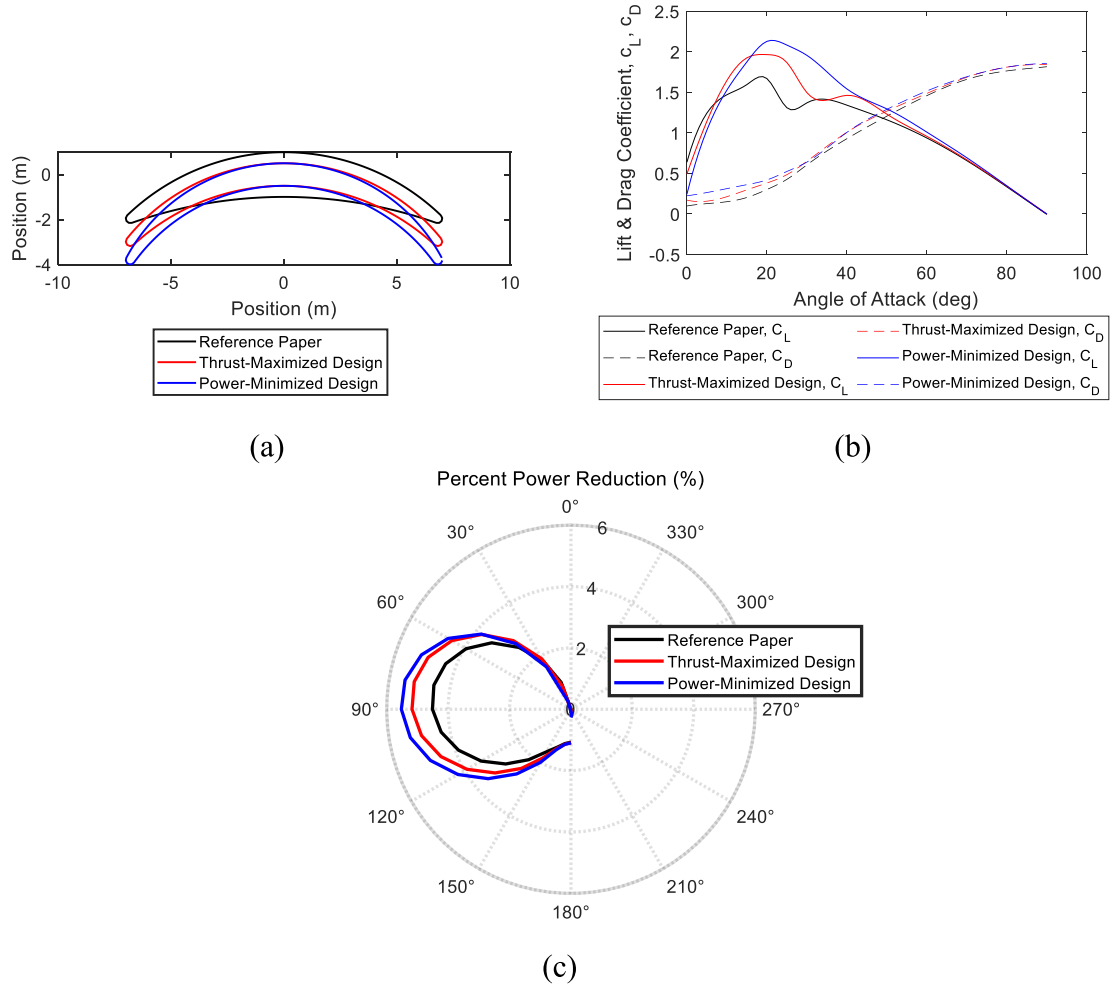


Fig. 9. Comparison of power-minimized, thrust maximized and reference airfoil in terms of (a) geometry, (b) lift and drag coefficient and (c) percent power reduction.

implemented is more efficient. Additionally, the proposed methodology for power-minimized design, allowing tailored design of airfoils for different ships and routes, is likely to be a better candidate to serve the industry as a tool for operational guidance.

Another focus of this study was to compare the differences between symmetric NACA-type and crescent-shaped airfoils. Crescent-shaped airfoils were observed to provide a greater increase in thrust and decreased in require propulsive power than NACA-type airfoils. Additionally, the leading and trailing edges of crescent-shaped airfoils are interchangeable, which entails their range of rotation is limited when compared to that of symmetric NACA airfoils, which, in turn, presents advantages in their implementation. Nonetheless, based on the airfoils considered in the present study, the maximum lift coefficients of crescent airfoils are larger than that of the considered symmetric NACA airfoil, and the drag behavior before stall shows significant differences. Hence, crescent-shaped airfoils, having larger drag coefficients before stall, become less efficient for head winds than NACA-type airfoils when the true wind angle is around 0–30°.

It is in order to point out that, in the current investigation, it was assumed that the sail is exposed to a steady inflow and that the angle of attack can be immediately adjusted. Fluctuations in wind speed and wind direction were discarded. In future studies, a robust crescent airfoil shape should be sought to have stable response in fluctuating inflow conditions. Having a robust airfoil provides advantages both in terms of performance and structural safety. Through the extension of the present

model into a three-dimensional unsteady simulation, further knowledge and design improvements can be obtained. Moreover, the performance of crescent airfoils can be improved by considering more sophisticated shapes.

5. Conclusion

The present study aimed to optimize the performance of crescent-shaped airfoils by shape optimization. The airfoil shape was parametrized, and the relation between performance and design was modeled by using surrogate models. The optimal designs which maximize the average thrust generated by the sail and minimize the average required propeller propulsive power were found. Finally, a comparison between different crescent-shaped airfoils and symmetric NACA airfoils was carried out.

The main findings of the present study are.

- Reducing thickness and increasing camber were shown to greatly improve the performance of the sail in terms of thrust maximization and power minimization.
- The average thrust generated by the thrust-maximized airfoil is about 12.3% larger than that of the reference crescent-shaped airfoil.
- The power-minimized airfoil improved the effectiveness of sails up to 22% with respect to the reference airfoil under the same environmental conditions for the target ship.

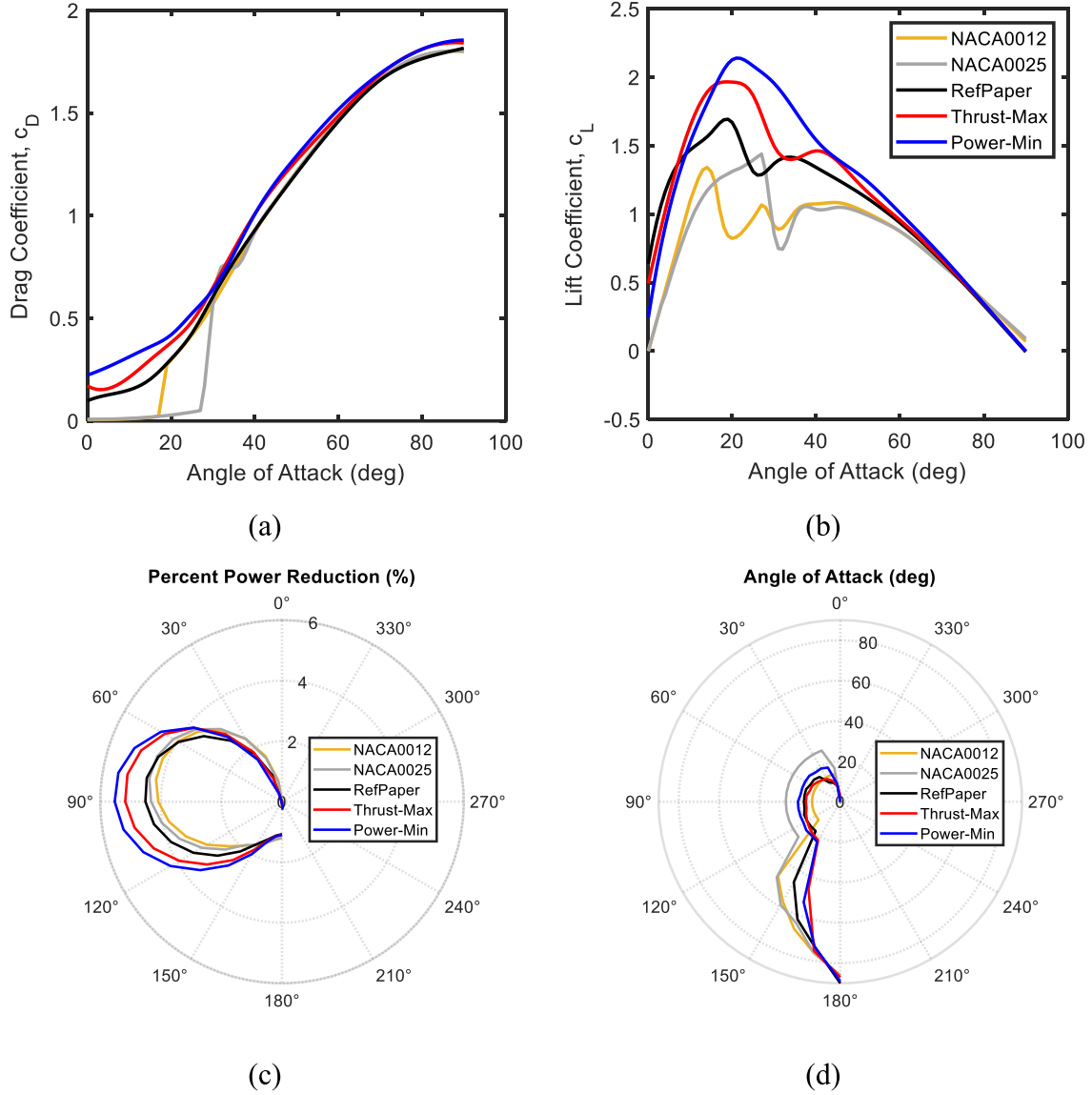


Fig. 10. Comparison of symmetric NACA airfoils, reference crescent airfoil and optimal crescent airfoils in terms of (a) drag and (b) lift coefficients, (c) percent power reduction and (d) optimal angle of attack depending on true wind direction.

- Including ship dynamics on the optimization stage of the sail design problem allows customizing the sails for specific ship and route through the proposed design optimization methodology.
- When the crescent-shaped airfoils and the symmetric NACA airfoils were compared, it was found that the crescent airfoils have a larger lift and drag coefficient in general, and they contribute more heavily than symmetric NACA airfoils to power reduction.

The present study focused on the energy performance of crescent shaped airfoils. Due to the adopted design of experiments study, design domain is limited within the variability of geometric parameters and there might be better solutions outside of the domain considered. Furthermore, three design parameters are used to describe the shape and the complexity of the optimal solution is also limited based on the prescribed three design variables. Increasing the number of variables increases the appearance of more complex shapes which in return requires higher computational cost and further performance improvements. In future studies, the marginal contribution of increasing shape complexity can be studied to determine the feasible complexity of the shape.

Although the power minimization approach performed better, the optimal results may exhibit differences depending on target ship and given route, which require additional calculations to find out the optimal design of sails for each condition. Another drawback of the study is that the evaluation of the airfoils was conducted considering only a fluid dynamics perspective and neglects structural integrity. Future studies are needed to ensure the safety and vibration aspects of sail design.

When multiple sail configurations were considered, the optimal results found for single sail and multiple sail may show some differences. In order to apply the proposed idea in multiple sail optimization, the layout and the design of sails should be investigated together to include the interaction effects.

Funding

This research was supported by REDAS: Fundamental Research Developing Association for Shipbuilding and Offshore, the Shipbuilders' Association of Japan.

CRedit authorship contribution statement

Cem Guzelbulut: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Timoteo Badalotti:** Writing – review & editing, Validation, Software, Investigation, Conceptualization. **Katsuyuki Suzuki:** Writing – review & editing, Supervision, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We would like to thank Maritime and Ocean Digital Engineering (MODE) Laboratory members at the University of Tokyo for their research advice.

Appendix. ASupplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.oceaneng.2024.119142>.

Appendix A

Table A.1

The coefficients used in Eq. (4).

	c_1	c_2	c_3	c_4	c_5	c_6	c_7
$C_{L,0}$	-1.59295	0.90464	5.18075	0.11902	-0.41213	-5.39578	-0.00485
$C_{L,10}$	1.29677	0.44347	2.01709	-0.02853	-0.21871	-2.83140	-0.00021
$C_{L,15}$	2.02693	0.13265	-0.24912	-0.01479	-0.11491	-0.74437	-0.00030
$C_{L,20}$	2.55835	-0.13773	-1.63040	-0.01623	-0.01155	0.11944	0.00009
$C_{L,25}$	3.35178	-0.00502	-5.16498	-0.08485	0.04325	3.98156	0.00300
$C_{L,30}$	2.69224	0.23789	-4.45132	-0.02003	-0.09651	3.36982	0.00063
$C_{L,40}$	1.85679	0.47259	-3.25471	0.01564	-0.24678	2.70661	-0.00061
$C_{L,50}$	1.55818	0.10170	-1.25954	-0.01192	-0.08530	0.88130	0.00025
$C_{L,60}$	1.14185	0.21489	-0.99938	-0.00502	-0.13256	0.77332	0.00004
$C_{L,70}$	0.77422	0.07574	-0.57117	0.00066	-0.05426	0.42246	-0.00015
$C_{L,80}$	0.35933	0.04133	-0.19118	0.00189	-0.02822	0.13208	-0.00012
$C_{L,90}$	0.00285	-0.00361	-0.00755	0.00002	0.00095	0.01156	0.00000
$C_{D,0}$	0.52098	0.02374	-1.34695	-0.01498	-0.01618	1.19199	0.00058
$C_{D,10}$	0.65682	0.02620	-1.47991	-0.02104	-0.02412	1.30488	0.00080
$C_{D,15}$	0.68613	0.04143	-1.30319	-0.02231	-0.02922	1.16387	0.00075
$C_{D,20}$	0.82632	0.14244	-1.32567	-0.04052	-0.07111	1.35816	0.00132
$C_{D,25}$	0.72848	0.34672	-0.61450	-0.03573	-0.19080	0.68900	0.00094
$C_{D,30}$	0.71127	0.22751	0.15280	-0.02328	-0.16324	-0.08994	0.00040
$C_{D,40}$	1.19855	-0.20287	1.10493	-0.04920	0.01778	-1.24669	0.00137
$C_{D,50}$	1.45348	-0.07205	-0.20991	-0.01183	-0.04614	0.04743	0.00013
$C_{D,60}$	1.62672	0.00825	-0.26376	-0.00597	-0.10197	0.08727	-0.00005
$C_{D,70}$	1.80495	-0.04602	-0.15952	-0.00500	-0.09477	-0.01609	-0.00006
$C_{D,80}$	1.92754	-0.05589	-0.09126	-0.00695	-0.10193	-0.06590	0.00005
$C_{D,90}$	2.00620	-0.05050	-0.00872	-0.01427	-0.10965	-0.19430	0.00034

Appendix B

Table B.1

The coefficients used in Eq. (9).

	d_1	d_2	d_3	d_4	d_5	d_6	d_7
VALUE	-3.5604	0.0612	0.4025	0.0230	0.1895	0.0120	-0.0005

References

- Altair Acusolve Software, 2023. AcuSolve (altair.com). <https://2021.help.altair.com/2021/hwsolvers/acusolve/index.htm>.
- Altair Hypermesh CFD Software, 2023. Streamlined CFD Simulation for Aerodynamics and Aeroacoustics | Altair® HyperMesh® CFD. <https://altair.com/hypermesh-cfd>.
- Cairns, J., Vezza, M., Green, R., MacVicar, D., 2021. Numerical optimisation of a ship wind-assisted propulsion system using blowing and suction over a range of wind conditions. Ocean Eng 240, 109903. <https://doi.org/10.1016/j.oceaneng.2021.109903>.
- Formosa, W., Sant, T., De Marco Muscat-Fenech, C., Figari, M., 2023. Wind-assisted ship propulsion of a Series 60 ship using a static kite sail. J. Mar. Sci. Eng. 11, 117. <https://doi.org/10.3390/jmse11010117>.
- Guzelbulut, C., Suzuki, K., 2023. Optimal design of rotor sails based on environmental conditions and cost. J. Mar. Sci. Eng. 12 (1), 31. <https://doi.org/10.3390/jmse12010031>.

- Guzelbulut, C., Sugimoto, T., Fujita, Y., Suzuki, K., 2024. Investigation of the efficiency of wind-assisted systems using model-based design approach. *J. Mar. Sci. Technol.* 29, 387–403. <https://doi.org/10.1007/s00773-024-00993-6>.
- IMO. Initial IMO strategy on reduction of GHG emissions from ships. Available online: <https://www.imo.org/en/OurWork/Environment/Pages/Vision-and-level-of-ambition-of-the-Initial-IMO-Strategy.aspx>. (Accessed 5 October 2023).
- Law, L.C., Foscoli, B., Mastorakos, E., Evans, S., 2021. A comparison of alternative fuels for shipping in terms of lifecycle energy and cost. *Energies* 14, 8502. <https://doi.org/10.3390/en14248502>.
- Lee, H., Jo, Y., Lee, D., Choi, S., 2016. Surrogate model based design optimization of multiple wing sails considering flow interaction effect. *Ocean Eng* 121, 422–436. <https://doi.org/10.1016/j.oceaneng.2016.05.051>.
- MOERI, 2008. KVLCC2 geometry and conditions. SIMMAN. FORCE Technology. Available online: http://www.simman2008.dk/KVLCC/KVLCC2/kvlcc2_geometry.html. (Accessed 27 June 2023).
- Nikmanesh, M., 2021. Sailing performance analysis using CFD simulations: a study on crescent shaped wing profiles. <https://odr.chalmers.se/handle/20.500.12380/304303>.
- Ogawa, A., Koyama, T., Kijima, K., 1977. MMG report-I, on the mathematical model of ship maneuvering. *Bull. Soc. Nav. Arch. Jpn* 575, 22–28 (In Japanese).
- Park, J.H., Choi, J.E., Chun, H.H., 2015. Hull-form optimization of KSUEZMAX to enhance resistance performance. *Int. J. Nav. Archit. Ocean Eng.* 7, 100–114. <https://doi.org/10.1515/ijnaoe-2015-0008>.
- Reche-Vilanova, M., Hansen, H., Bingham, H.B., 2021. Performance prediction program for wind-assisted cargo ships. *J. Sailing Technol* 6 (1), 91–117. <https://doi.org/10.5957/jst/2021.6.1.91>.
- Roy, R.K., 1990. *A Primer on the Taguchi Method*, first ed. Society of Manufacturing Engineers.
- Tadros, M., Ventura, M., Guedes Soares, C., 2021. Design of propeller series optimizing fuel consumption and propeller efficiency. *J. Mar. Sci. Eng.* 9, 1226. <https://doi.org/10.3390/jmse9111226>.
- Tadros, M., Ventura, M., Guedes Soares, C., 2022. Towards fuel consumption reduction based on the optimum contra-rotating propeller. *J. Mar. Sci. Eng.* 10, 1657. <https://doi.org/10.3390/jmse10111657>.
- Tillig, F., Ringsberg, J.W., 2019. A 4 DOF simulation model developed for fuel consumption prediction of ships at sea. *Ships Offshore Struct.* 14, 112–120. <https://doi.org/10.1080/17445302.2018.1559912>.
- Tomos, B.A.D., Stamford, L., Welfle, A., Larkin, A., 2024. Decarbonising international shipping – a life cycle perspective on alternative fuel options. *Energy Convers. Manag.* 299, 117848. <https://doi.org/10.1016/j.enconman.2023.117848>.
- Viola, I.M., Sacher, M., Xu, J., Wang, F., 2015. A numerical method for the design of ships with wind-assisted propulsion. *Ocean Eng* 105, 33–42. <https://doi.org/10.1016/j.oceaneng.2015.06.009>.
- Yasukawa, H., Yoshimura, Y., 2015. Introduction of MMG standard method for ship maneuvering predictions. *J. Mar. Sci. Technol.* 20, 37–52. <https://doi.org/10.1007/s00773-014-0293-y>.
- Zhu, H., Yao, H.D., Thies, F., Ringsberg, J.W., Ramne, B., 2023. Propulsive performance of a rigid wingsail with crescent-shaped profiles. *Ocean Eng* 285, 115349. <https://doi.org/10.1016/j.oceaneng.2023.115349>.